Oracle 2 types of databases –

1) **Standalone** – Single server
2) **RAC** (Real Application Cluster) – 2 types – RAC database and RAC Onenode database

- **Standalone without ASM:** install database on O.S File System/ OS storage – DEVDB – oraclelab1
- **Standalone with ASM:** install database on ASM File System/ ASM storage – restart server - TESTDB – oraclelab2

1) Standalone without ASM -> oraclelab1.rar (Link #1) (OS file system / OS storage) - DEVEB

oraclelab1.localdomain.com (Linux VM 7.9)

1. Setup oraclelab1 server - 192.168.0.151
2. Download 19c Oracle Software and Install
3. Create Database -> DEVDB

2) Standalone with ASM -> oraclelab2.rar (Link #1) (ASM file system / ASM storage) - TESTDB

oraclelab2.localdomain.com (Linux VM 7.9)

1. Setup oraclelab2 server - 192.168.0.152
2. Download 19c ASM Software and Install and Setup ASM storage
3. Download 19c Oracle Software and Install
4. Create Database -> TESTDB

1) Standalone without ASM -> oraclelab1.rar (Link #1) (OS file system / OS storage) - DEVEB

oraclelab1.localdomain.com (Linux VM 7.9)

1. Setup oraclelab1 server - 192.168.0.151
2. Download 19c Oracle Software and Install
3. Create Database -> DEVDB

1. Setup oraclelab1 server -

Download oraclelab1.rar file from Link #1 >>> Ready_VMs_Standalone (folder)

https://drive.google.com/drive/folders/183dcnQtLtEp8mS3MX3VvCyiKjrpNGicT?usp=sharing



Download and install winrar from link #2 software to unzip oraclelab1.rar >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing



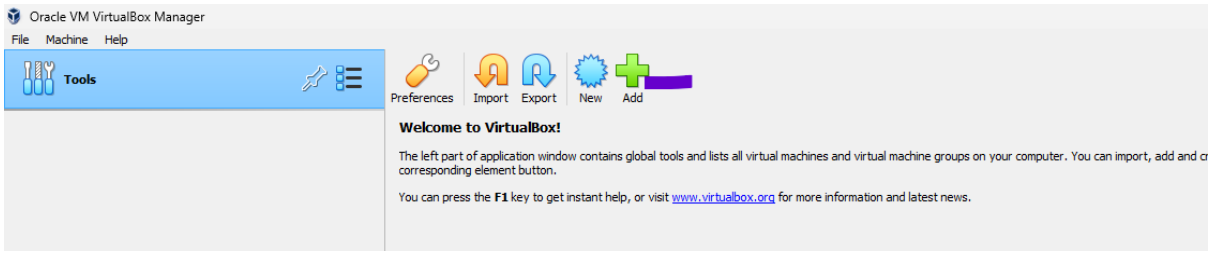
Download and Install Oracle Virtualbox from Link# 2 and setup oraclelab1 >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing

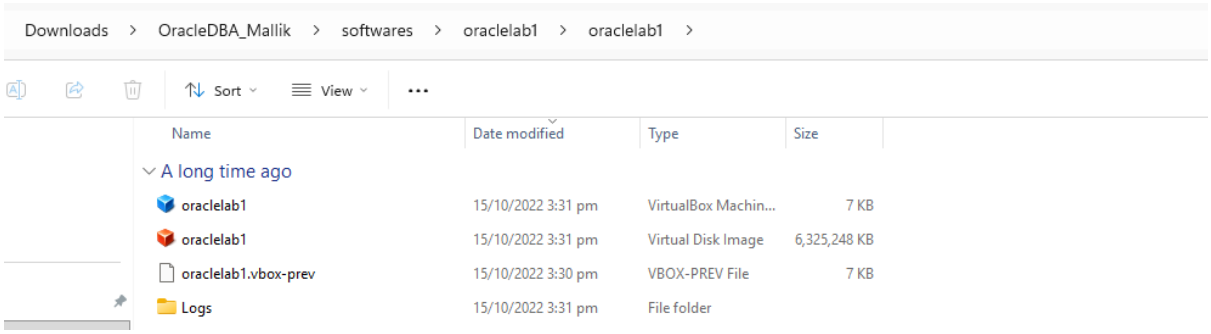


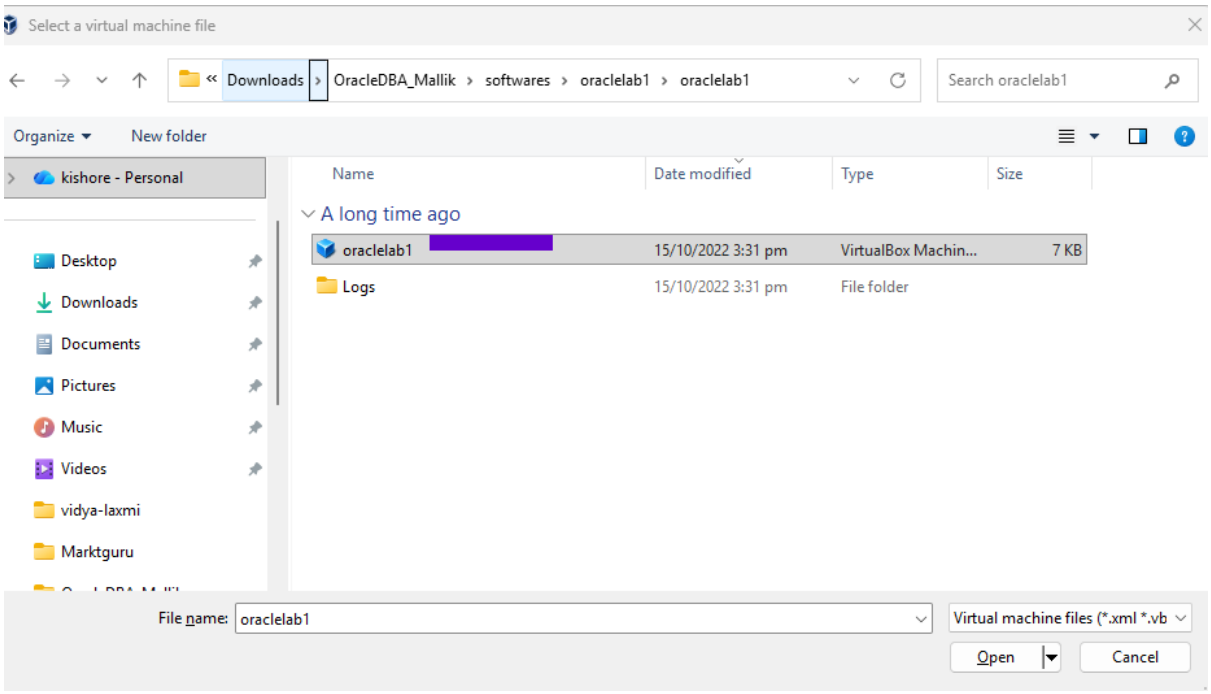
Post installation of VirtualBox, open – add

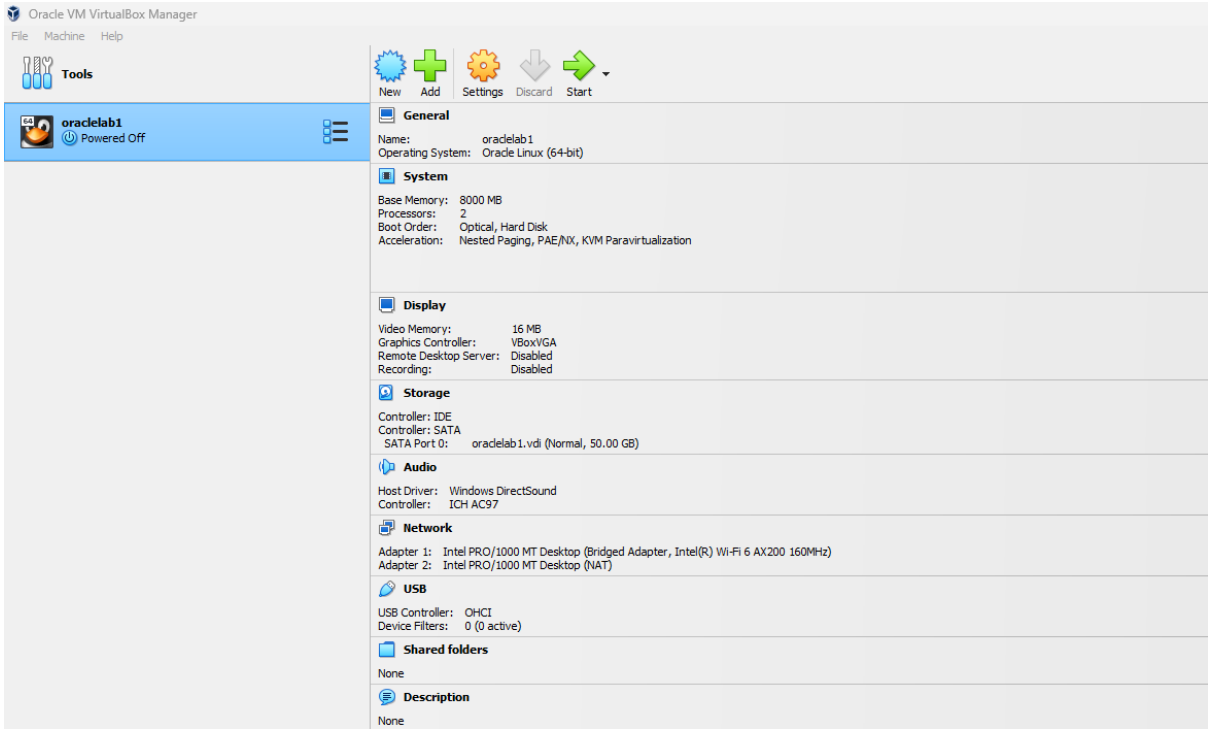
- Setup oraclelab1 - add oraclelab1 to oracle virtual box



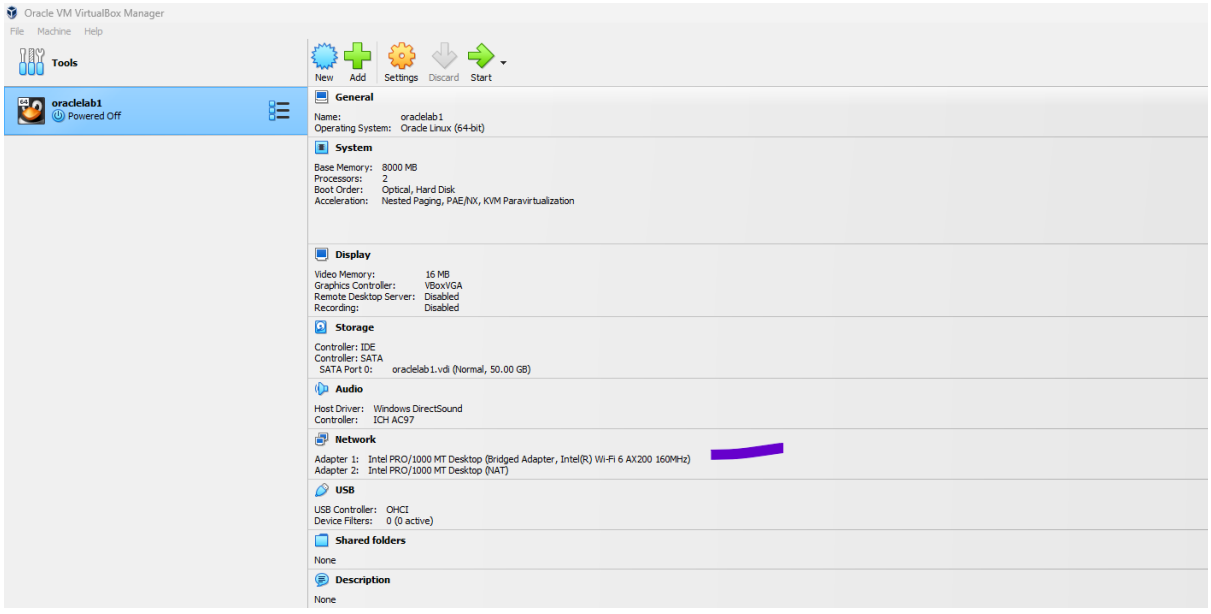
After extracting oraclelab1.rar







<u>Network setup</u> (Its done by network admin and they will assign IP address)
Remove the existing network settings and add our own laptop IP -



Both make it as NAT

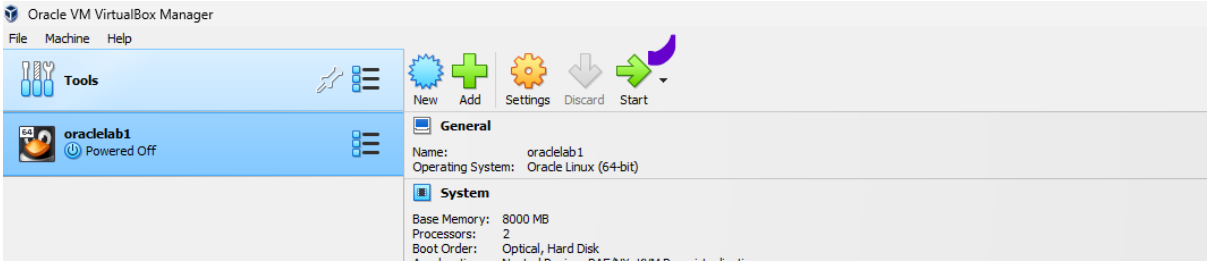


Then again change them back to Bridge adapter

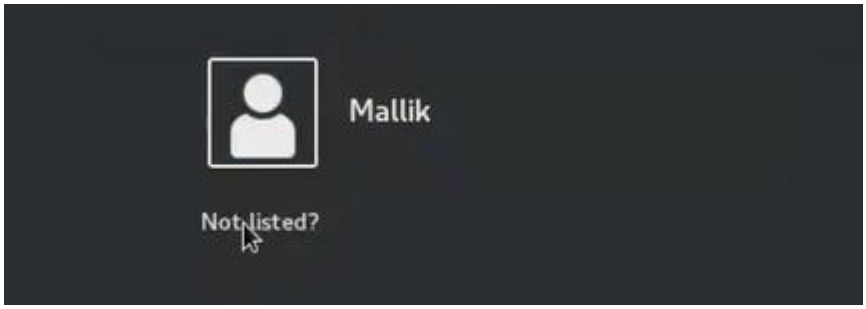


The moment we change to bridged adapter it will pick up the new laptop IP details –

Now, start the VM –



Don't click on Malik user – click on not listed



Root – pwd: Mallik@123



Next, IP configuration –

From local laptop get these details -

cmd> ipconfig

cmd> ipconfig /all

```
C:\Users\Dell>ipconfig

Windows IP Configuration
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6955:aa9b:b95c:b22c%20
   IPv4 Address. . . . . . . . . . . : 192.168.1.107
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.254

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

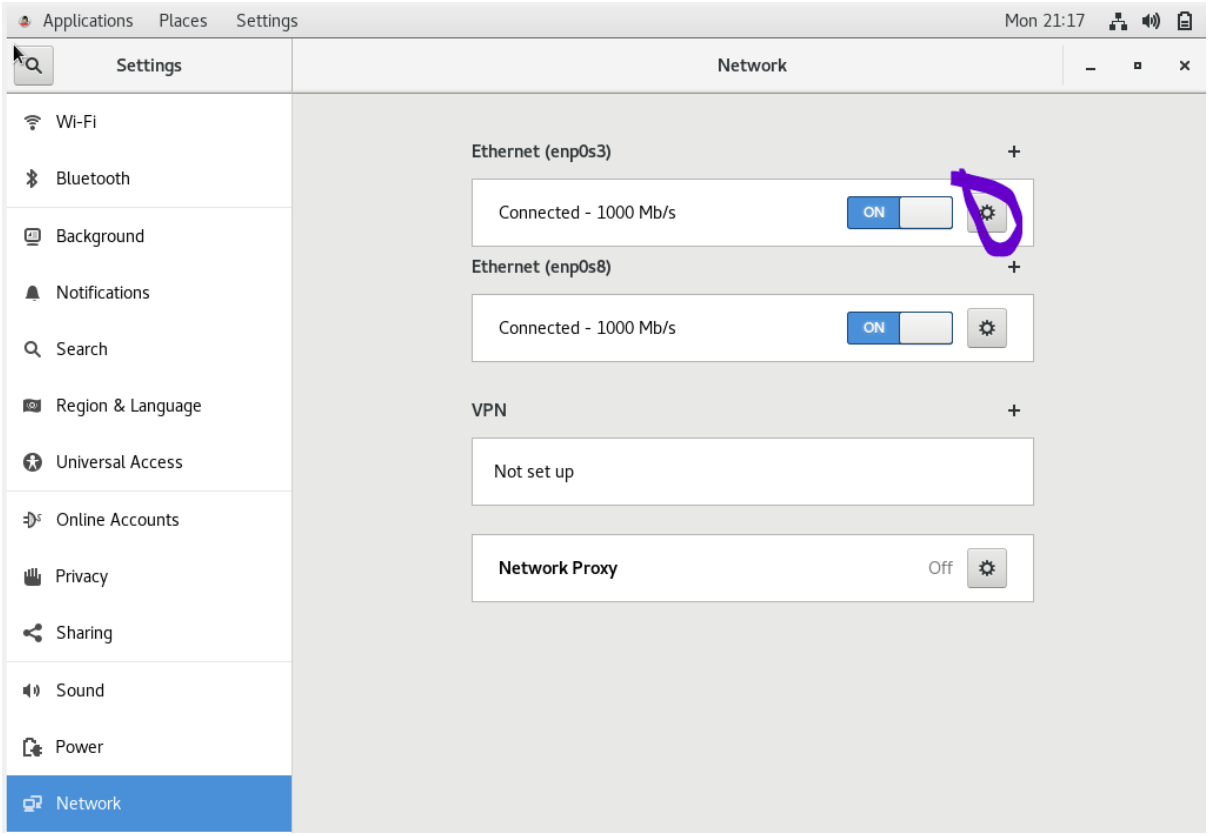
C:\Users\Dell>
```

- get laptop/desktop IP address -> 192.168.1.107
- Subnet - 255.255.255.0
- gateway -> 192.168.1.254
- Get DNS IP address from laptop/desktop -> 192.168.1.254

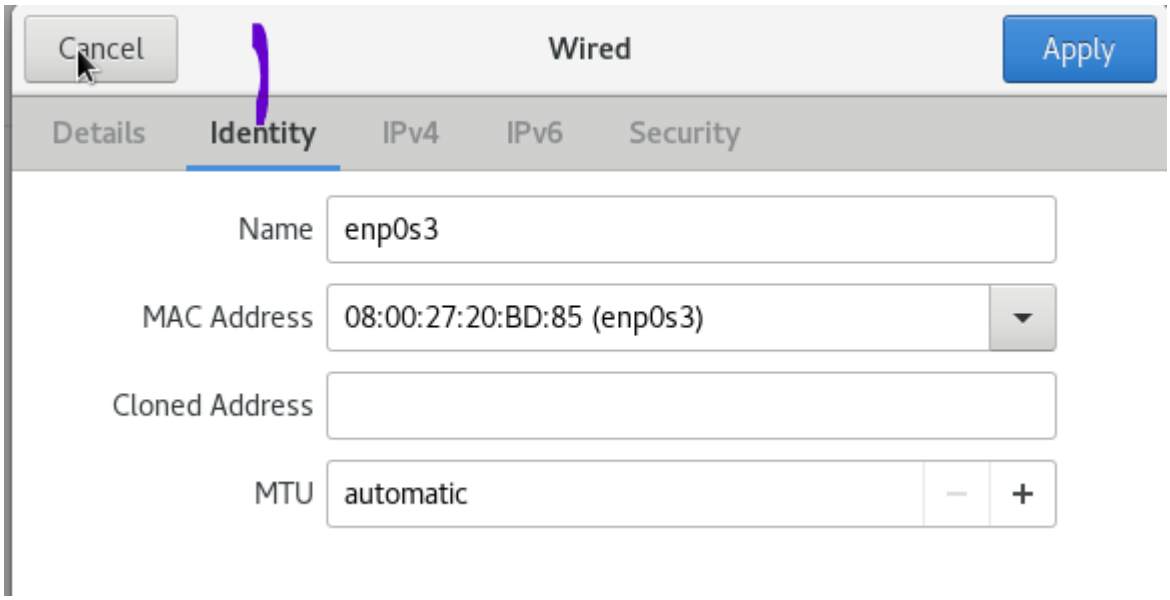
- Assign IP address to VM oraclelab1 in the same range of your laptop/desktop IP address -> **192.168.1.151** (10 to 240)

Assign this new IP – VirtualBox

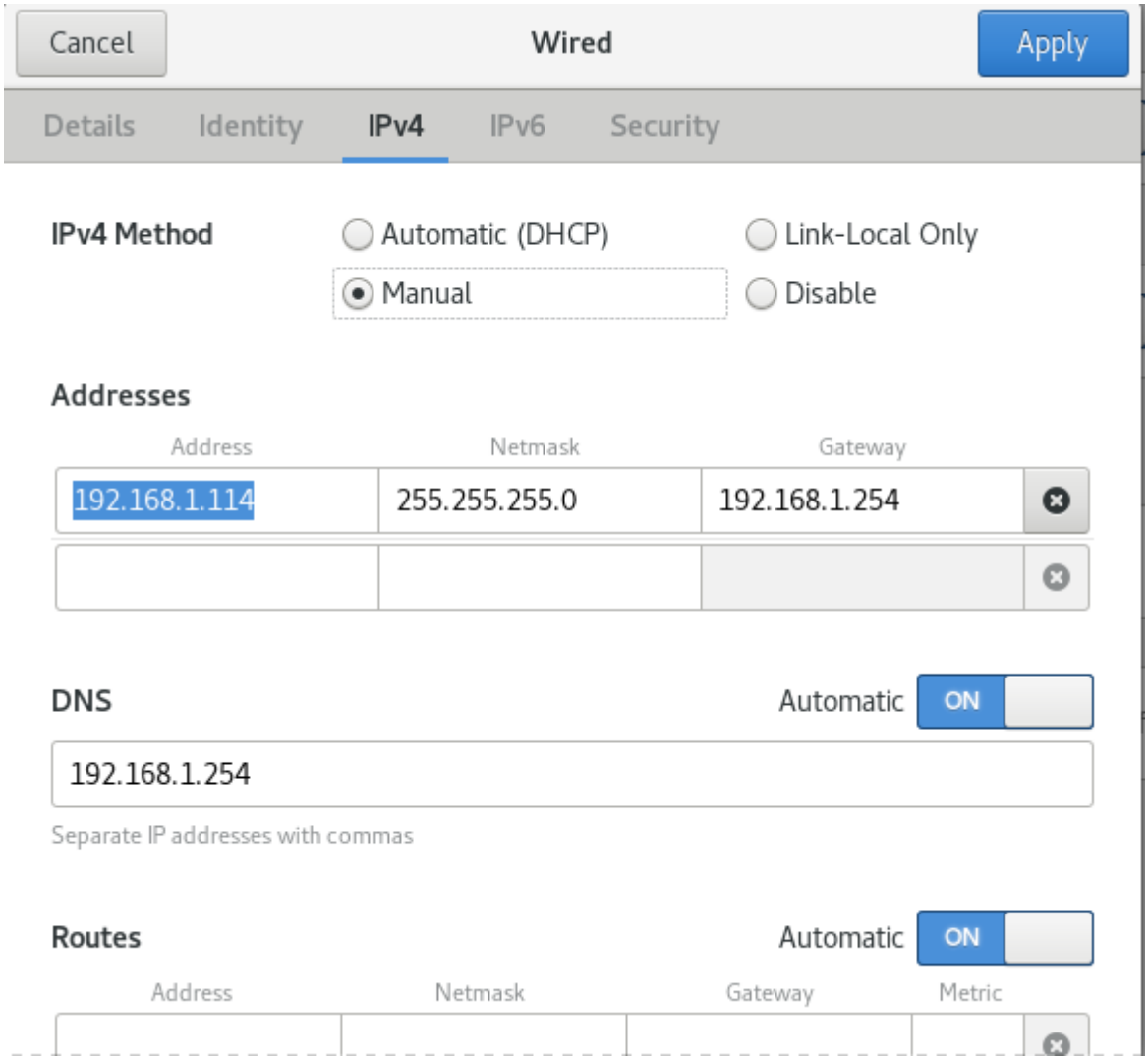
Applications -> system tools -> settings -> Network


Applications
Places
Settings
Mon 21:17
Settings
Network
Wi-Fi
Bluetooth
Background
Notifications
Search
Region & Language
Universal Access
Online Accounts
Privacy
Sharing
Sound
Power
Network
Ethernet (enp0s3)
Connected - 1000 Mb/s
ON
Ethernet (enp0s8)
Connected - 1000 Mb/s
ON
VPN
Not set up
Network Proxy
Off

Settings ->


Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
Name
enp0s3
MAC Address
08:00:27:20:BD:85 (enp0s3)
Cloned Address
MTU
automatic

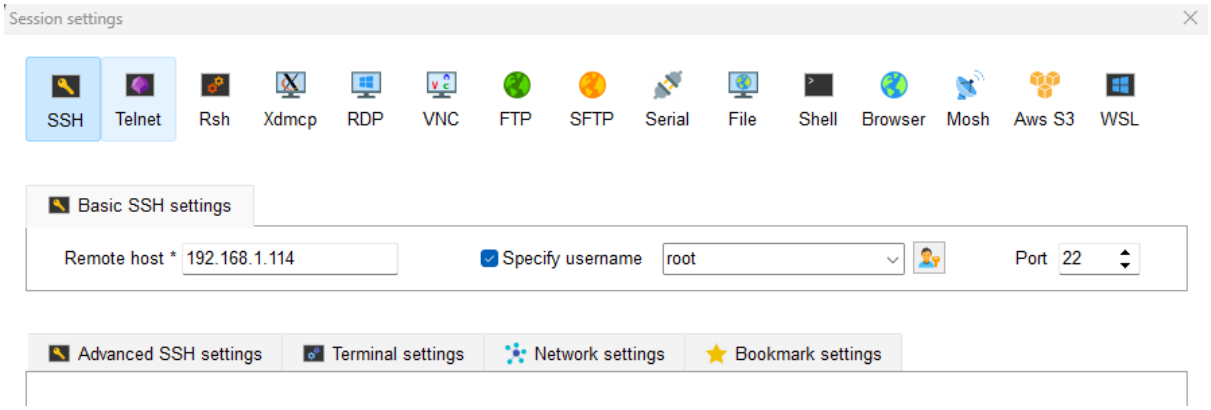
IPV4 tab -

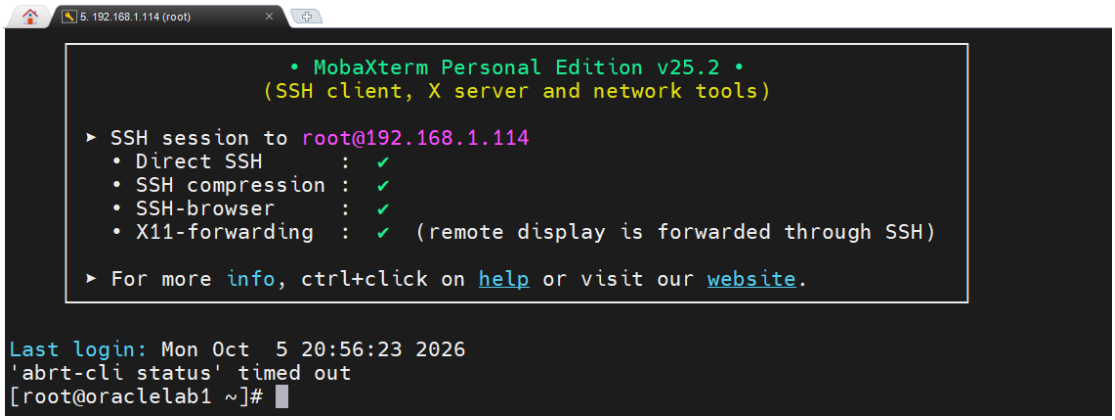


- Access oraclelab1 (192.168.1.151) from putty / mobaxterm, putty / mobaxterm software can be downloaded from link #2 and Install on your laptop >>> Other Tools for DBAs (folder)

https://drive.google.com/drive/folders/1mh5j3hBMad0cCpV7mhph2mmyAGYfwx3U?usp=sharing

From MobaXterm – connect –

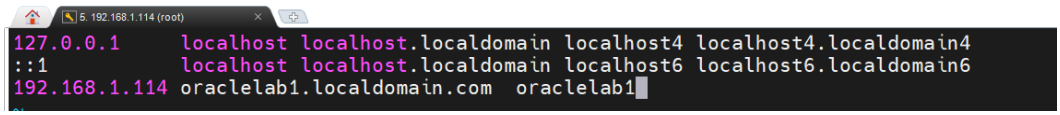




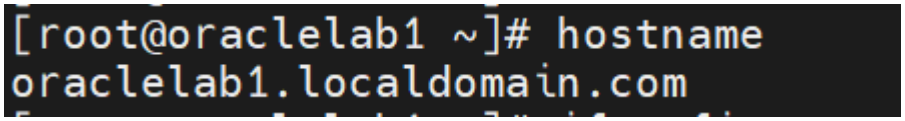
- add IP address (192.168.1.101) into vi /etc/hosts file

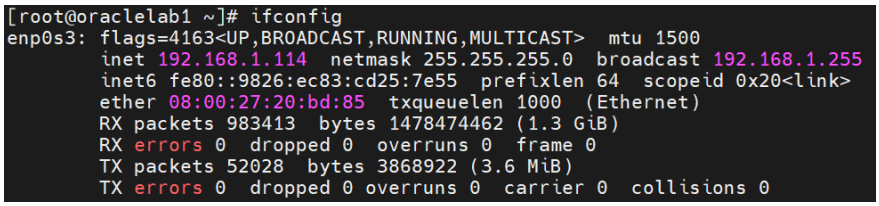
192.168.0.151 oraclelab1.localdomain.com oraclelab1





validate



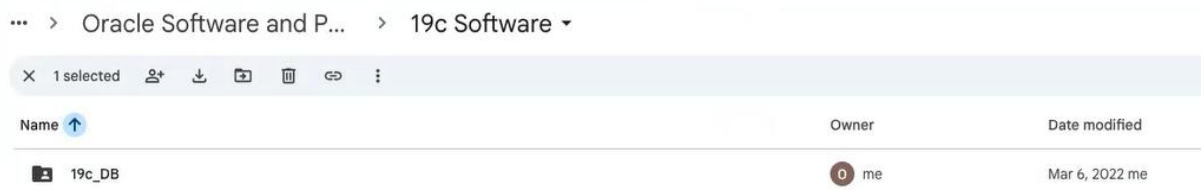


2. Download and Install Oracle Home/Oracle Software

Download 19c Oracle Software and Install ->

- ❖ Oracle Home (or) Oracle Software (or) DB Software (or) DB Home -> all are same

a. Download DB software (19c) from https://edelivery.oracle.com (Oracle Database 19c 19.3.0.0.0) or - Download it from Google drive Link #2 >>> Oracle Software and Patches >>> 19c Software >>> 19c_DB



Below one – GI means Grid Infrastructure which is for ASM

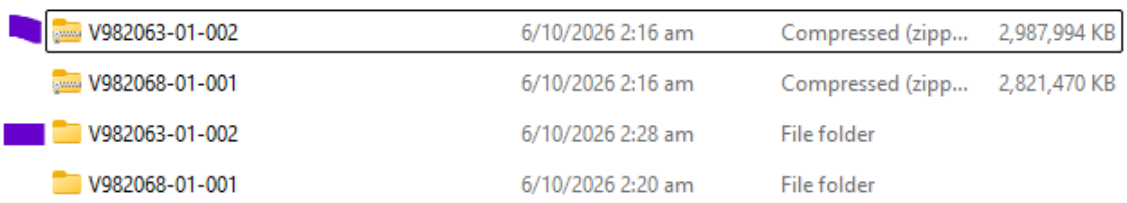


-

https://drive.google.com/drive/folders/1Oa9c0lhRhOMaEUB9UQiaaKp0BmqZNYWn

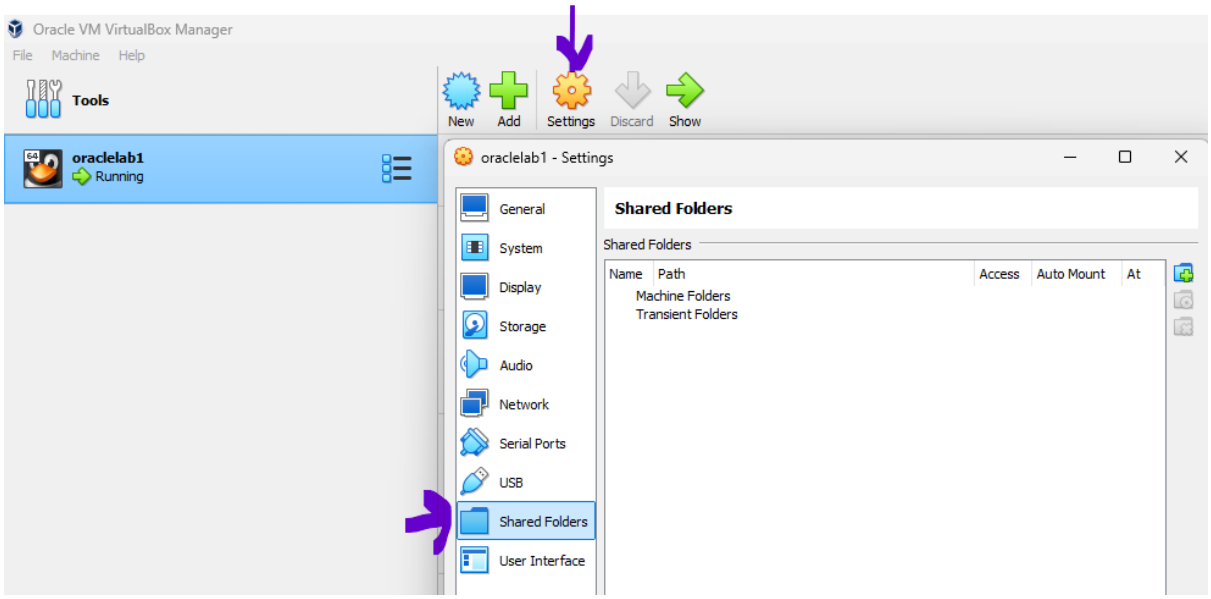
- Oracle Database 19c 19.3.0.0.0 (Oracle Database In-Memory, Oracle Database Vault)

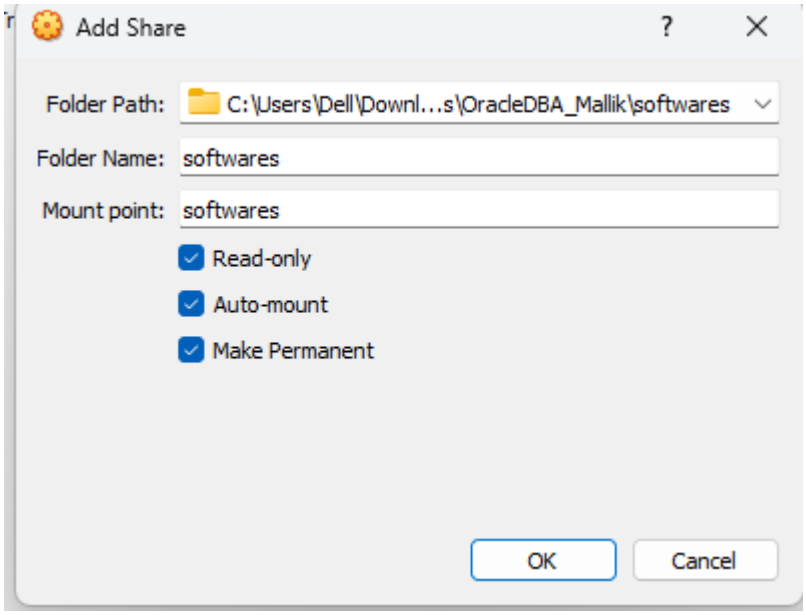
- DB - V982063-01.zip



b. Install Oracle Software / Oracle Home / DB Home / DB software

- Copy DB SW from lapttop/Desktop to VM (oraclelab1)
  - Use winscp
    - WinSCP is software or tool download from Link #2
  - Shared folder (preferred)





We can see the shared folder in linux



```
[root@oraclelab1 ~]# df -h
Filesystem           Size  Used Avail Use% Mounted on
devtmpfs             3.7G     0  3.7G   0% /dev
tmpfs                3.7G     0  3.7G   0% /dev/shm
tmpfs                3.7G  9.4M  3.7G   1% /run
tmpfs                3.7G     0  3.7G   0% /sys/fs/cgroup
/dev/mapper/ol-root   34G  7.7G   27G  23% /
/dev/sda1           1014M  231M  784M  23% /boot
softwares            476G  319G  158G  67% /media/sf_softwares
tmpfs                749M   24K  749M   1% /run/user/0
[root@oraclelab1 ~]# cd ^C
[root@oraclelab1 ~]# cd /media/sf_softwares
[root@oraclelab1 sf_softwares]# ls
oraclelab1      V982063-01-002      V982068-01-001.zip
oraclelab1.rar  V982063-01-002.zip  VirtualBox-7.0.20-163906-Win.exe
oraclelab2.rar  V982068-01-001      winrar-x64-621.exe
[root@oraclelab1 sf_softwares]#
```

c. Directory creation & SW unzip

- mkdir -p /u01/app/oracle/product/19.0.0.0/dbhome_1 (this is the directory where we install Oracle19c software)

- unzip /media/sf_Software/DB_19c_Software/V982063-01.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1

(or)

- unzip V982063-01.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1

```
[root@oraclelab1 sf_softwares]# mkdir -p /u01/app/oracle/product/19.0.0.0/dbhome_1
[root@oraclelab1 sf_softwares]# unzip V982063-01-002.zip -d /u01/app/oracle/product/19.0.0.0/dbhome_1
```

```
tcjt.so
  /u01/app/oracle/product/19.0.0.0/dbhome_1/jdk/jre/bin/ControlPanel -> jcontrol
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/admin/lfclasses.bin -> ../../javavm/jdk/jdk8/admin/lf
classes.bin
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/cacerts -> ../../../javavm/jdk/jdk8/lib/
security/cacerts
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/sunjce_provider.jar -> ../../javavm/jdk/jdk8/lib/
sunjce_provider.jar
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/README.txt -> ../../../javavm/jdk/jdk8/l
ib/security/README.txt
  /u01/app/oracle/product/19.0.0.0/dbhome_1/javavm/lib/security/java.security -> ../../../javavm/jdk/jdk
8/lib/security/java.security
  /u01/app/oracle/product/19.0.0.0/dbhome_1/jdk/jre/lib/amd64/server/libjsig.so -> ../libjsig.so
[root@oraclelab1 sf_softwares]#
```

d. Download and install pre-requisite RPMs – all 12c,19c,21c etc.

- yum install oracle* --skip-broken

(It will set all kernel parameters and it will install all pre-requisite rpms and set all shared memory setting and create all oracle user & groups)

```
[root@oraclelab1 V982063-01-002]# cd /u01/app/oracle/product/19.0.0.0/dbhome_1/
[root@oraclelab1 dbhome_1]# ls
addnode     data         env.ora        jlib     OPatch   plsql        root.sh.old.1  srvm
apex        dbjava       has            ldap     opmn     precomp      runInstaller   suptools
assistants  dbs          hs             lib      oracore  QOpatch      schagent.conf  ucp
bin         deinstall    install        md       ord      R            sdk            usm
clone       demo         instantclient  mgw      ords     racg         slax           utl
crs         diagnostics  inventory      network  oss      rdbms        sqldeveloper   wwg
css         dmu          javavm         nls      oui      relnotes     sqlj           xdk
ctx         drdaas       jdbc           odbc     owm      root.sh      sqlpatch
cv          dv           jdk            olap     perl     root.sh.old  sqlplus
[root@oraclelab1 dbhome_1]# yum install oracle* --skip-broken
Loaded plugins: langpacks, ulninfo
Package oracle-logos-70.7.0-1.0.7.el7.noarch already installed and latest version
Package oracle-spacewalk-client-release-el7 is obsoleted by oracle-linux-manager-client-release-el7, tr
ing to install oracle-linux-manager-client-release-el7-1.0-2.el7.x86_64 instead
Package oracle-spacewalk-server-release-el7 is obsoleted by oracle-linux-manager-server-release-el7, tr
ing to install oracle-linux-manager-server-release-el7-1.0-4.el7.x86_64 instead

Dependency Installed:
  ksh.x86_64 0:20120801-144.0.1.el7_9                libaio-devel.x86_64 0:0.3.109-13.el7

Updated:
  oraclelinux-release.x86_64 7:7.9-1.0.13.el7        oraclelinux-release-el7.x86_64 0:1.0-17.el7

Skipped (dependency problems):
  oracle-linux-manager-client-release-el7.x86_64 0:1.0-2.el7

Complete!
[root@oraclelab1 dbhome 1]#
```

/*this step- optional – skip- if you want only specific 19c then install as below*/

- yum -y install oracle-database-preinstall-19c

- yum -y install oracle-database-preinstall-21c

https://docs.oracle.com/en/database/oracle/oracle-database/19/ladbi/running-rpm-packages-to-install-oracle-database.html

- Optional (In case if you installed above RPM)

e. User & group create (Not required) – as part of rpm installation, this will create users and groups

groupadd dba

groupadd oinstall

useradd -g oinstall -G oinstall,dba oracle

```
[root@oraclelab1 dbhome_1]# id oracle
uid=54321(oracle) gid=54321(oinstall) groups=54321(oinstall),54322(dba)
[root@oraclelab1 dbhome_1]#
```

So, with this single command everything will be created - - **yum install oracle* --skip-broken**

Advantages of oracle pre-requisite rpm installation

- all kernel parameter are set

- all the required rpms are auto installed

- all the oracle user and groups are auto created

f. Change permission

#chown -R oracle:oinstall /u01

#Oracle user already created but we don't know the password so set it

- passwd oracle (set password) - Mallik@123

```
[root@oraclelab1 ~]# chown -R oracle:oinstall /u01
[root@oraclelab1 ~]# passwd oracle
Changing password for user oracle.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@oraclelab1 ~]#
```

3. Database SW Installation / Oracle SW Installation / Oracle Home Installation (from GUI screen as oracle user)

Login with oracle user

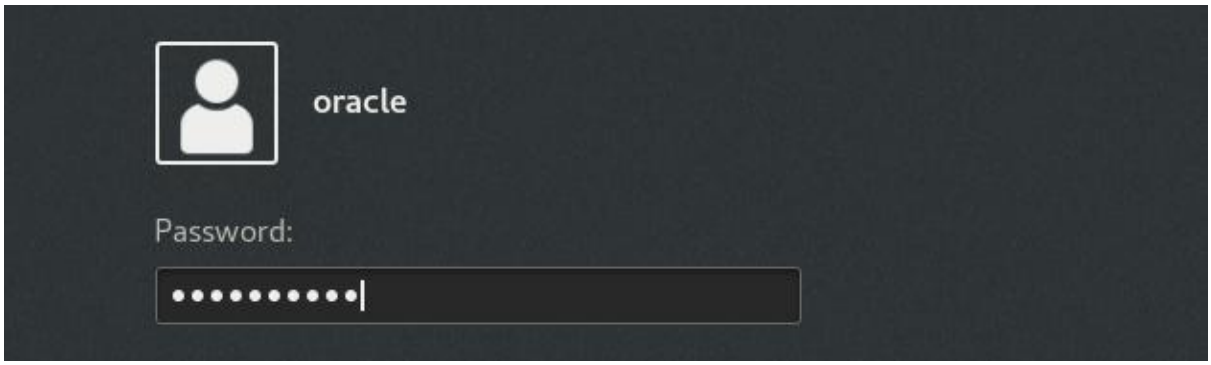
$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1

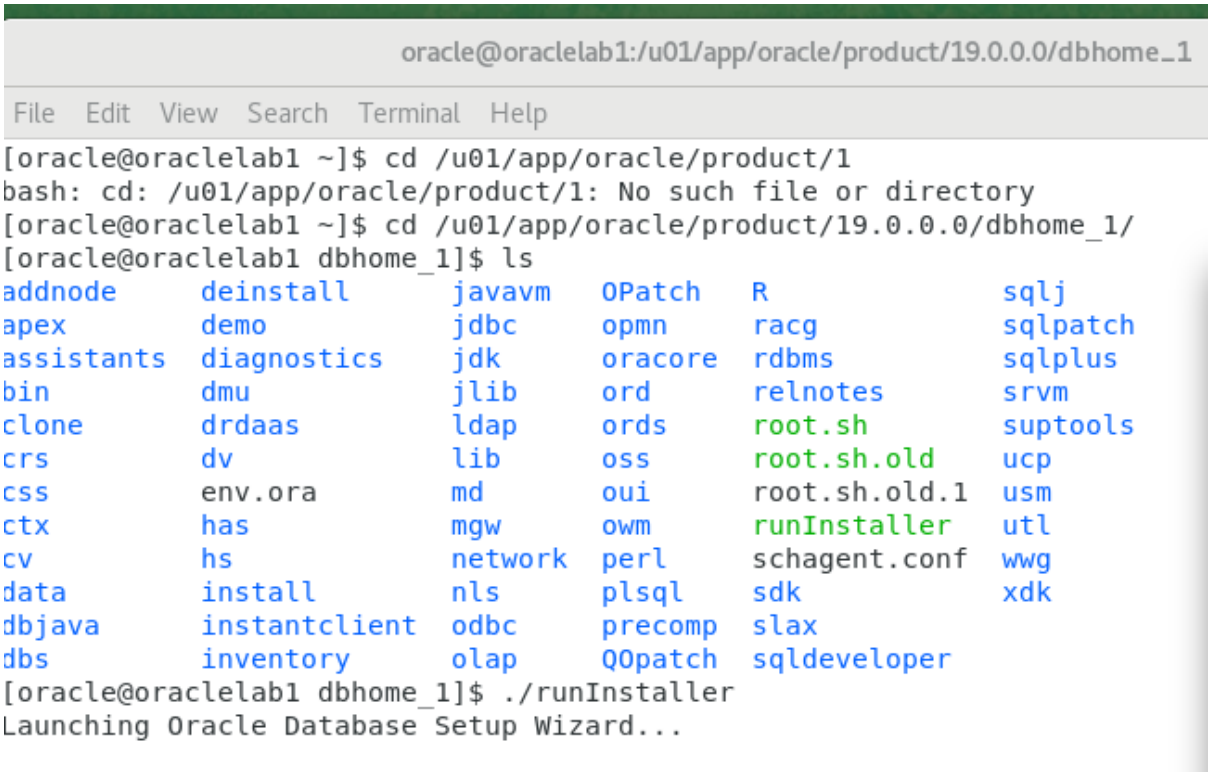
$ ./runInstaller

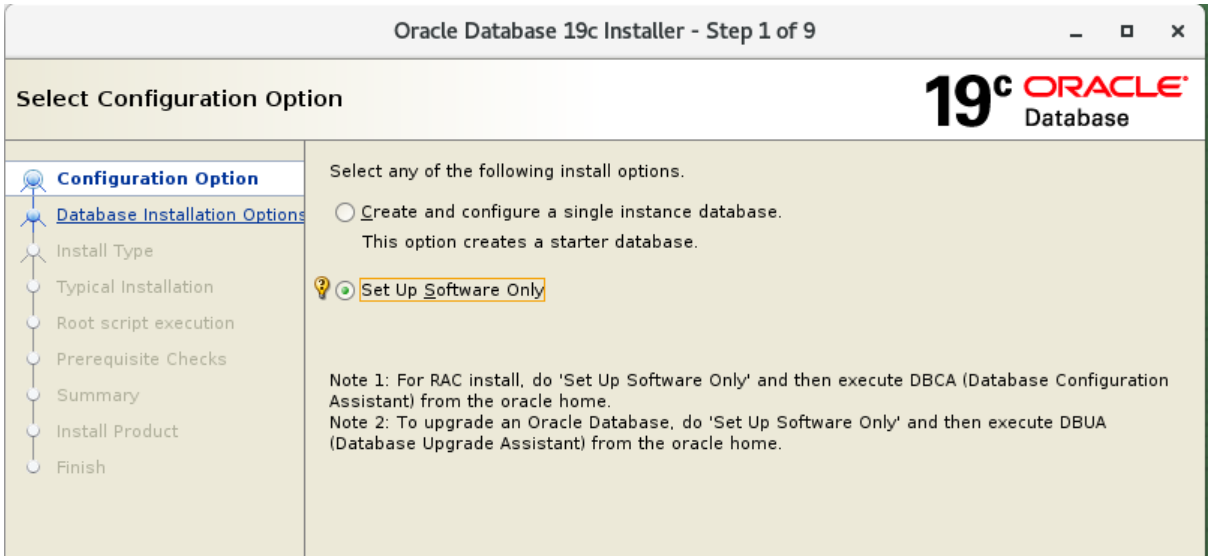
```
[oracle@oraclelab1 ~]$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/
[oracle@oraclelab1 dbhome_1]$ ./runInstaller
ERROR: Unable to verify the graphical display setup. This application requires X displ
 xdpyinfo exist under PATH variable.

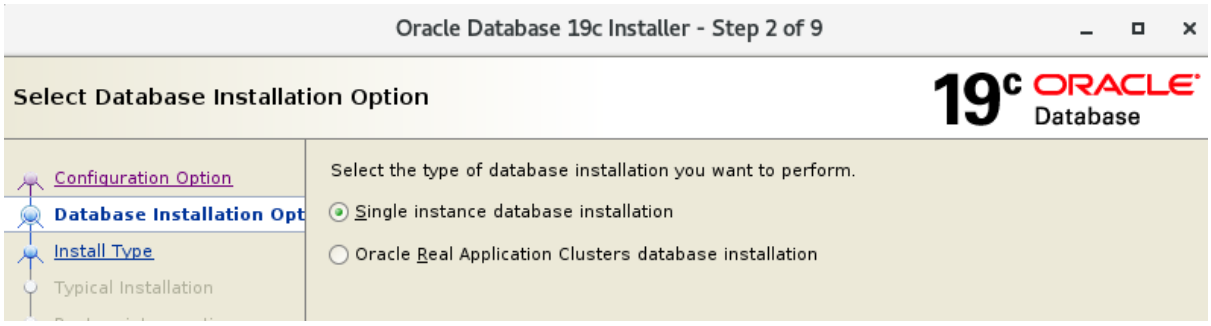
No X11 DISPLAY variable was set, but this program performed an operation which require
[oracle@oraclelab1 dbhome_1]$ xhost +
xhost:  unable to open display ""
[oracle@oraclelab1 dbhome_1]$ su - root
Password:
Last login: Fri Oct  9 06:35:33 IST 2026 on pts/0
[root@oraclelab1 ~]# su - oracle
Last login: Fri Oct  9 06:37:38 IST 2026 on pts/0
[oracle@oraclelab1 ~]$ xhost +
xhost:  unable to open display ""
[oracle@oraclelab1 ~]$
```

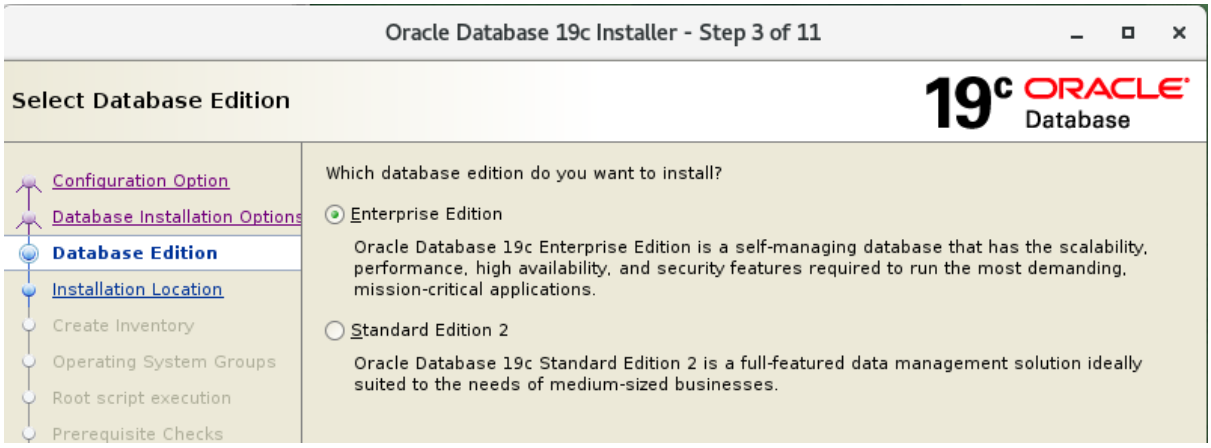
Error – we need to directly login with oracle user (not root user) – logout and login with oracle user

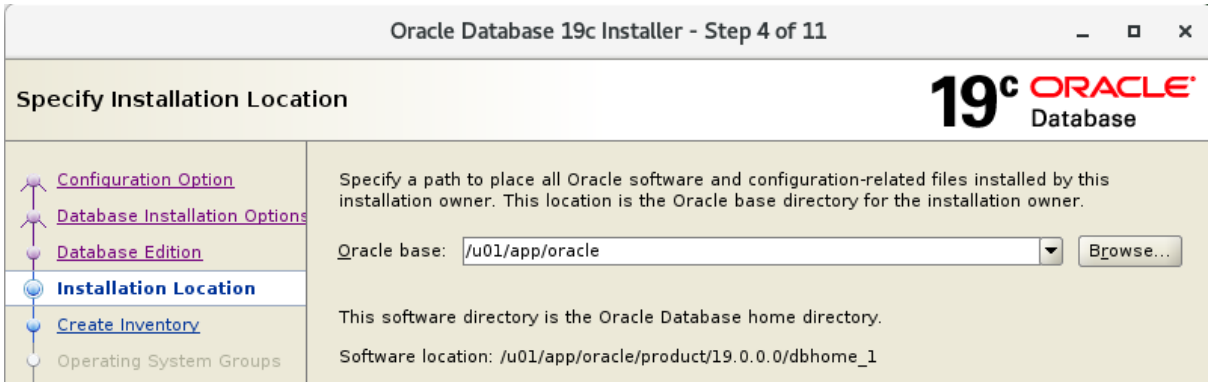


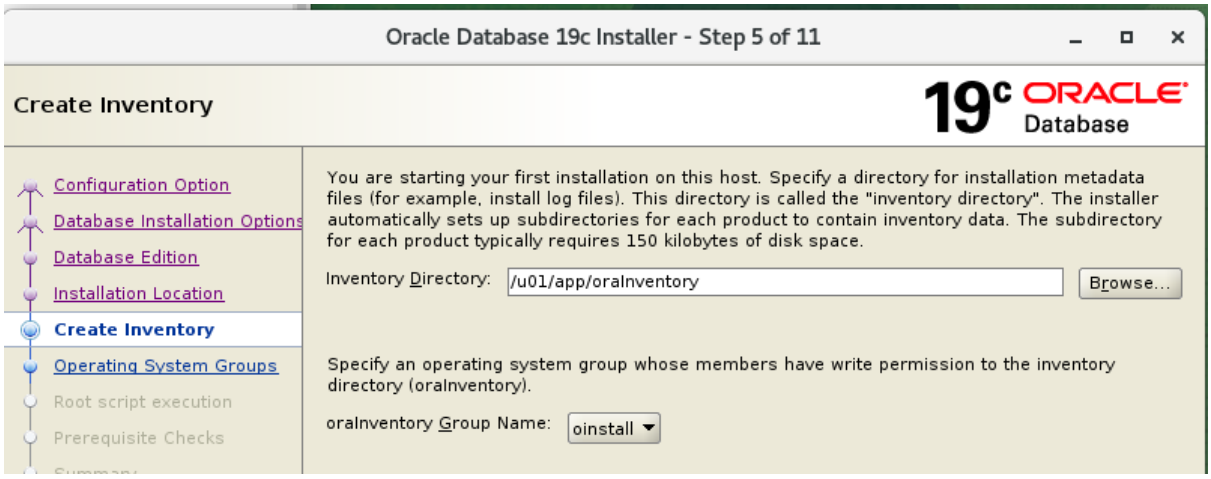

oracle@oraclelab1:/u01/app/oracle/product/19.0.0.0/dbhome_1
File Edit View Search Terminal Help
[oracle@oraclelab1 ~]$ cd /u01/app/oracle/product/1
bash: cd: /u01/app/oracle/product/1: No such file or directory
[oracle@oraclelab1 ~]$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/
[oracle@oraclelab1 dbhome_1]$ ls
addnode deinstall javavm OPatch R sqlj
apex demo jdbc opmn racg sqlpatch
assistants diagnostics jdk oracore rdbms sqlplus
bin dmu jlib ord relnotes srvm
clone drdaas ldap ords root.sh suptools
crs dv lib oss root.sh.old ucp
css env.ora md oui root.sh.old.1 usm
ctx has mgw owm runInstaller utl
cv hs network perl schagent.conf wwg
data install nls plsql sdk xdk
dbjava instantclient odbc precomp slax
dbs inventory olap QOpatch sqldeveloper
[oracle@oraclelab1 dbhome_1]$ ./runInstaller
Launching Oracle Database Setup Wizard...

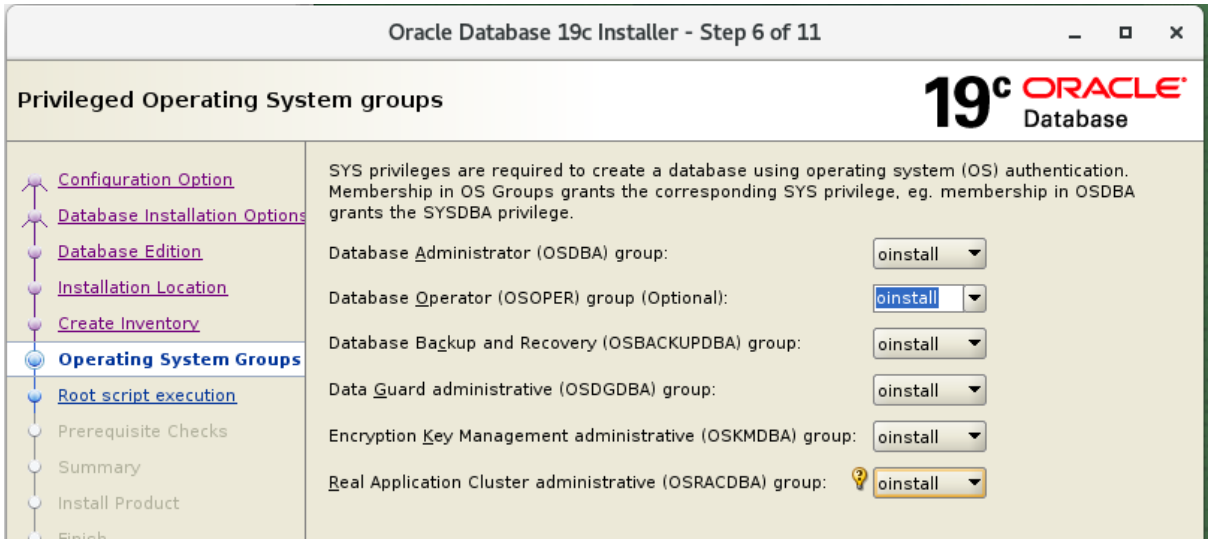

Oracle Database 19c Installer - Step 1 of 9
Select Configuration Option
19c ORACLE Database
Configuration Option
Database Installation Options
Install Type
Typical Installation
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Select any of the following install options.
Create and configure a single instance database.
This option creates a starter database.
Set Up Software Only
Note 1: For RAC install, do 'Set Up Software Only' and then execute DBCA (Database Configuration Assistant) from the oracle home.
Note 2: To upgrade an Oracle Database, do 'Set Up Software Only' and then execute DBUA (Database Upgrade Assistant) from the oracle home.

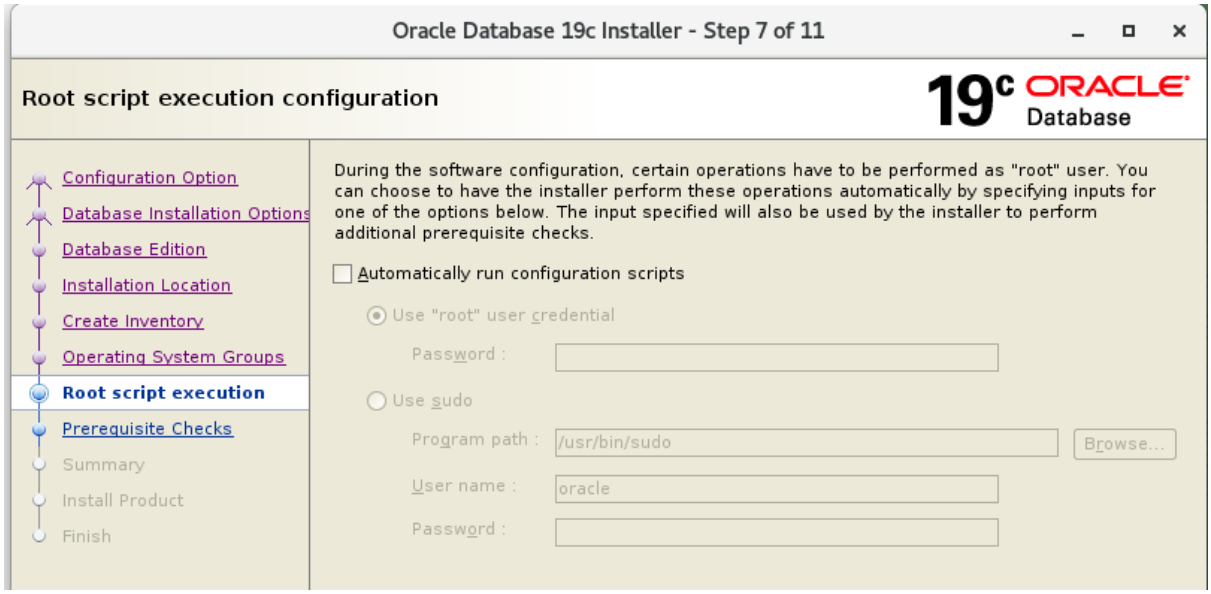

Oracle Database 19c Installer - Step 2 of 9
Select Database Installation Option
19c ORACLE Database
Configuration Option
Database Installation Opt
Install Type
Typical Installation
Select the type of database installation you want to perform.
Single instance database installation
Oracle Real Application Clusters database installation

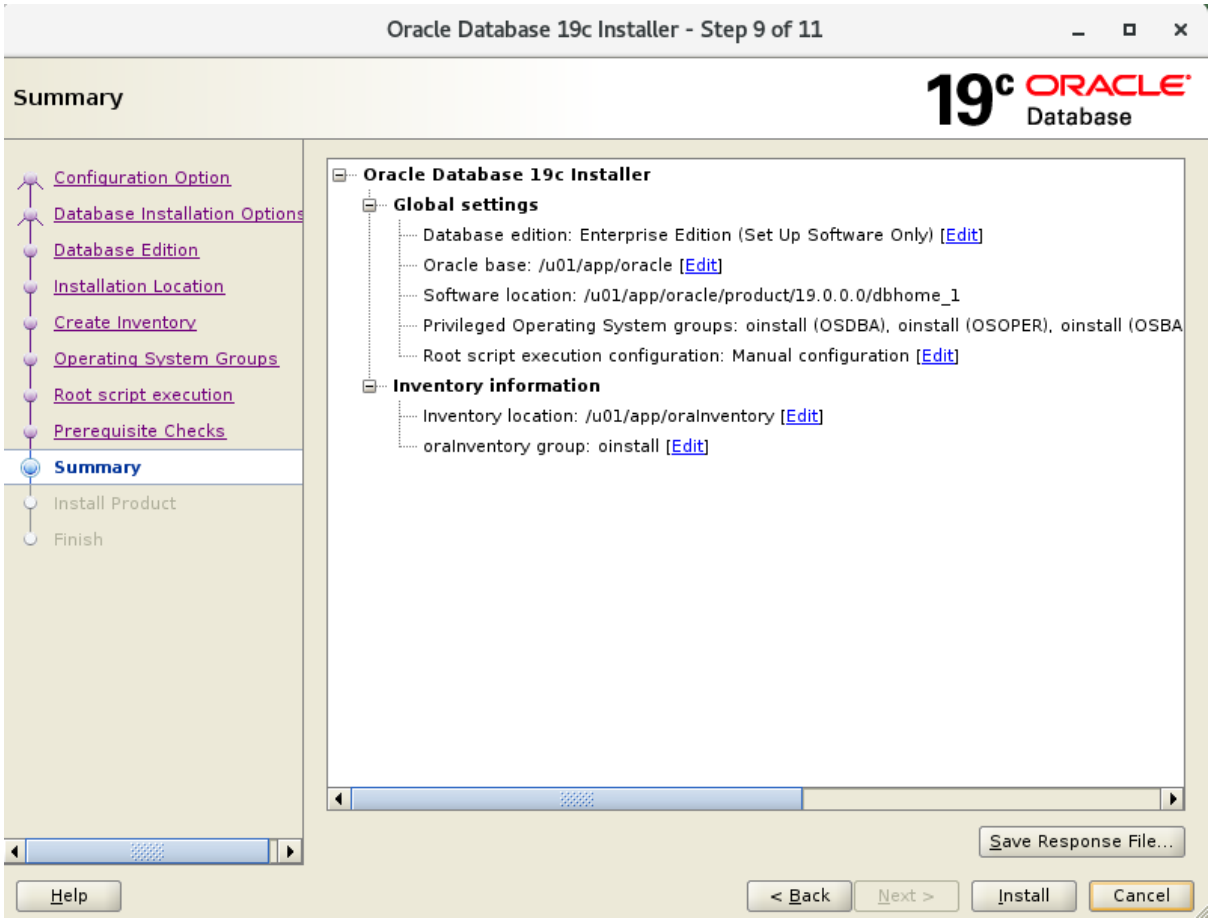

Oracle Database 19c Installer - Step 3 of 11
Select Database Edition
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Which database edition do you want to install?
Enterprise Edition
Oracle Database 19c Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.
Standard Edition 2
Oracle Database 19c Standard Edition 2 is a full-featured data management solution ideally suited to the needs of medium-sized businesses.

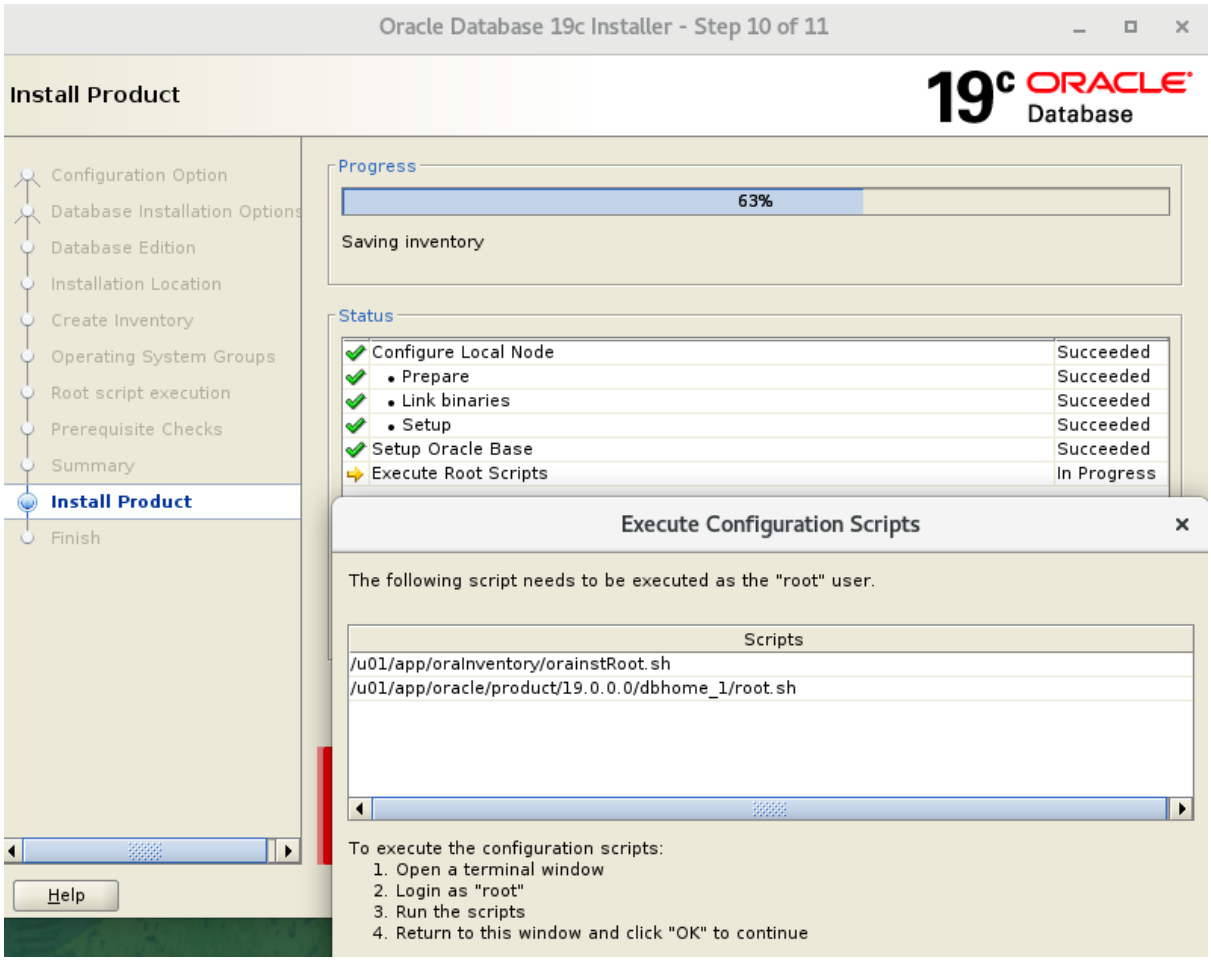

Oracle Database 19c Installer - Step 4 of 11
Specify Installation Location
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.
Oracle base: /u01/app/oracle
Browse...
This software directory is the Oracle Database home directory.
Software location: /u01/app/oracle/product/19.0.0.0/dbhome_1


Oracle Database 19c Installer - Step 5 of 11
Create Inventory
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.
Inventory Directory: /u01/app/oraInventory
Browse...
Specify an operating system group whose members have write permission to the inventory directory (oraInventory).
oraInventory Group Name: oinstall


Oracle Database 19c Installer - Step 6 of 11
Privileged Operating System groups
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Summary
Install Product
SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.
Database Administrator (OSDBA) group: oinstall
Database Operator (OSOPER) group (Optional): oinstall
Database Backup and Recovery (OSBACKUPDBA) group: oinstall
Data Guard administrative (OSDGDBA) group: oinstall
Encryption Key Management administrative (OSKMDBA) group: oinstall
Real Application Cluster administrative (OSRACDBA) group: oinstall


Oracle Database 19c Installer - Step 7 of 11
Root script execution configuration
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.
Automatically run configuration scripts
Use "root" user credential
Password :
Use sudo
Program path : /usr/bin/sudo
Browse...
User name : oracle
Password :


Oracle Database 19c Installer - Step 9 of 11
Summary
19c ORACLE Database
Configuration Option
Database Installation Options
Database Edition
Installation Location
Create Inventory
Operating System Groups
Root script execution
Prerequisite Checks
Summary
Install Product
Finish
Oracle Database 19c Installer
Global settings
Database edition: Enterprise Edition (Set Up Software Only) [Edit]
Oracle base: /u01/app/oracle [Edit]
Software location: /u01/app/oracle/product/19.0.0.0/dbhome_1
Privileged Operating System groups: oinstall (OSDBA), oinstall (OSOPER), oinstall (OSBA
Root script execution configuration: Manual configuration [Edit]
Inventory information
Inventory location: /u01/app/oraInventory [Edit]
oraInventory group: oinstall [Edit]
Save Response File...
Help
< Back
Next >
Install
Cancel



Run these scripts

```
[oracle@oraclelab1 dbhome_1]$ su - root
Password:
Last login: Fri Oct  9 06:40:48 IST 2026 on pts/0
[root@oraclelab1 ~]# /u01/app/oraInventory/orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oinstall.
The execution of the script is complete.
[root@oraclelab1 ~]# /u01/app/oracle/product/19.0.0.0/dbhome_1/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME=  /u01/app/oracle/product/19.0.0.0/dbhome_1

Enter the full pathname of the local bin directory: [/usr/local/bin]:
```

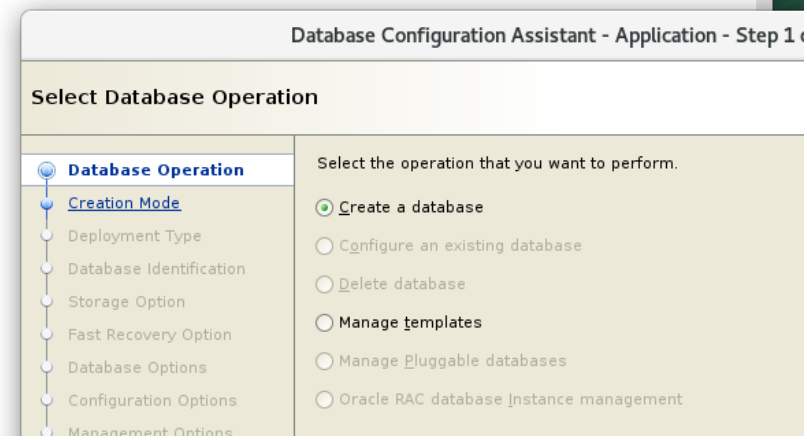
Press enter -

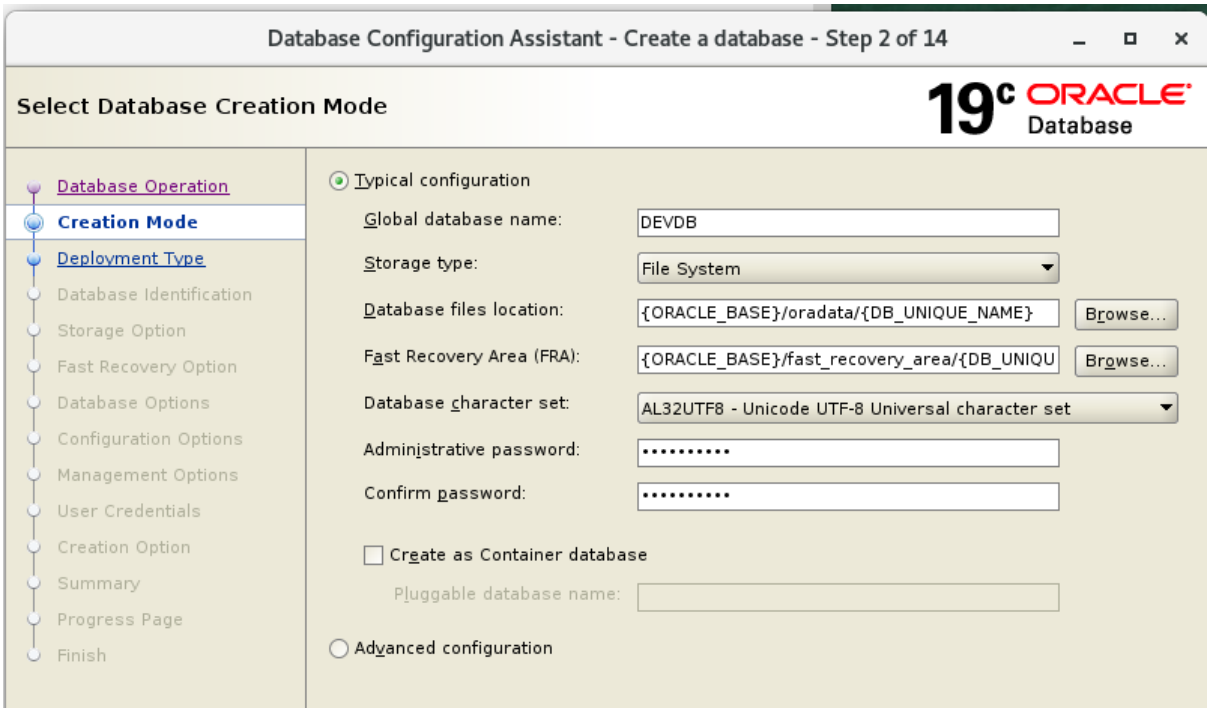
*** Oracle HOME installation is completed ***

3. Create Database -> DEVDB with oracle user

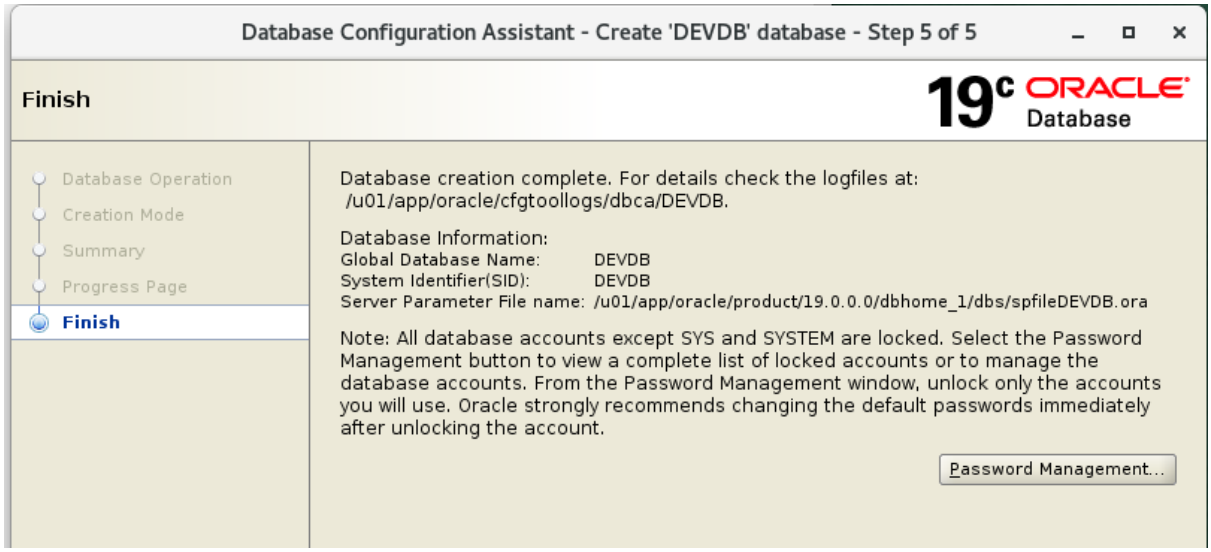
```
$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/bin (from GUI screen as oracle user)
$ ./dbca
```

```
[oracle@oraclelab1 bin]$ cd /u01/app/oracle/product/19.0.0.0/dbhome_1/bin/
[oracle@oraclelab1 bin]$ ls -l dbca
-rwxr-x--- 1 oracle oinstall 7749 Oct  9 06:52 dbca
[oracle@oraclelab1 bin]$ ./dbca
```





Mallik@123




```
[root@oraclelab1 ~]# ps -ef | grep smon
oracle    8179     1  0 07:10 ?        00:00:00 ora_smon_DEVDB
root      8550  5104  0 07:12 pts/1    00:00:00 grep --color=auto smon
[root@oraclelab1 ~]#
```

********** DONE **********